

A PRACTICAL FIELD BOOKLET

Electricity from Heat, Copper, Water & Motion

How to convert fire, metal, flowing water, and magnets into useful electric current — without pretending energy can appear from nowhere.

*Low-voltage educational projects only.
Not a license to wire a house, tap the grid, or build boilers.*

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1. The rule that never breaks

Electricity is not created from nothing. It is converted from some other form of energy: heat in a fire, chemical energy in two different metals, the motion of water or a spinning magnet, or light on a solar cell. “Free electricity” slogans usually mean one of two things: a real conversion that costs fuel, parts, or labor — or a scam that pretends a magnet or a copper pipe can run forever.

This booklet stays on the first kind. You can light an LED, charge a small battery bank slowly, or run a radio for a while. You cannot lawfully or safely replace the utility with a campfire and scrap copper, and you cannot beat conservation of energy.

Keep this sentence

Energy in = energy out, minus losses as heat and friction. If a diagram shows more power coming out than going in, the diagram is wrong.

Useful conversions in this book: heat difference → voltage (Seebeck). Motion of a magnet past copper wire → voltage (Faraday). Two metals in water or salt → a weak battery. Falling water turning a wheel → a spinning generator.

2. Safety before any spark

Every method below is meant for **low voltage**: typically under 12 volts DC, small currents, and projects you can unplug with your hands. House wiring, generators tied into a wall outlet, and high-pressure steam are outside this book on purpose.

Do not do these

Do not tap power lines or a neighbor's meter. Do not backfeed a homemade generator into a wall socket — that can kill utility workers. Do not build a pressurized steam boiler from cans. Do not electrolyze water expecting “free hydrogen fuel.” Splitting water costs more electricity than the gas will ever return. Do not mix strong acids. Keep open flame outdoors or in a fireplace, never unattended, and never near gasoline or propane leaks.

Minimum kit

- Multimeter (volts DC, milliamps).
- Safety glasses, dry gloves, fire extinguisher or water bucket for campfire work.
- Insulated copper wire, not house mains cable.
- A small LED, a 1.5–5 V USB power-bank circuit only if you know it is isolated, or a cheap USB meter.
- A metal pan or brick barrier between flame and electronics.

3. Fire and heat — thermoelectric generation

A temperature difference across certain materials produces a voltage. That is the Seebeck effect. Camping “stove fans” and some survival chargers use it. The practical part is a **Peltier module** (TEC1-12706 is common): one face hot, one face cold.

What you need

- One or more Peltier / thermoelectric cooler modules.
- A heat source: wood stove, candle cluster, or a small outdoor fire under a pan of sand — not a blowtorch on the module.
- A cold side: a metal mug of ice water, a heat sink in a stream, or a thick aluminum plate sitting in a wet cloth.
- Thermal paste, two metal plates, and short wires to an LED or a 5 V boost board rated for the module.

How to set it up

1. Sandwich the module between two flat metal plates. The printed side usually faces the cold plate; check the seller diagram. 2. Put the hot plate on a pan above coals or on a wood-stove surface that is warm, not glowing. 3. Keep the cold plate actually cold — ice water or running water. The voltage comes from the *difference*, not from how huge the fire is. 4. Measure open-circuit voltage first. A single module often gives about 1–5 V depending on the gradient. 5. Several modules in series raise voltage; in parallel they raise current. 6. Feed a small LED, a USB boost converter designed for thermoelectric input, or a low-voltage hobby motor.

Fire limits

Peltier modules melt internally if the hot face exceeds roughly 150–180 °C. Use a pan of sand or a thick plate as a buffer. Never drop the module in the flames. Output is modest: think phone trickle or LED lantern, not a refrigerator.

A cruder heat method is a thermocouple chain: twist copper wire to constantan or even salvaged thermocouple wire, heat one junction in a candle, leave the other at room temperature. You get millivolts per junction — enough to demonstrate the effect, not enough to live on.

Steam engines and turbines convert fire to motion, then motion to electricity. They are real industrial technology. Homemade boilers explode. This booklet does not give boiler, pressure-vessel, or turbine build steps. If you want fire → motion at hobby scale, look at a commercially made Stirling-engine kit with a generator pulley — still small watts, still not free.

4. Copper — coils, magnets, and batteries

A. Faraday generator (the honest copper path)

Moving a magnet past a coil of copper wire induces voltage. That is how almost every generator on Earth works, from a bicycle dynamo to a hydro

plant. Copper does not make energy. It is the path for current while mechanical energy turns the magnet.

Parts: enamel-coated magnet wire (30–200 turns on a plastic spool or PVC), strong neodymium magnets, a crank, bicycle wheel, or small water wheel, a diode bridge if you want DC, and an LED.

Build: Wind a tight coil. Mount magnets on a spinning disk so they sweep past the coil face (alternating north/south around the disk works better). Spin by hand or by a belt from a bicycle. More turns raise voltage; stronger magnets and higher speed raise both voltage and current. Rectify with four diodes if you want to charge a small capacitor or NiMH cell through a proper charge circuit.

Copper pipe myth

A copper tube with a magnet sliding inside looks magical because the magnet falls slowly. That is eddy-current braking — the magnet's energy is turning into heat in the copper. It is not a generator you can plug a house into.

B. Copper–zinc wet cell (chemistry, not magic)

Two different metals in an electrolyte make a battery. Copper and zinc (or a galvanized nail) in salt water, vinegar, or lemon juice is the school version of a voltaic cell.

- Clean copper sheet or pipe as the positive plate.
- Zinc (galvanized nail, zinc anode, or the case of a spent carbon-zinc battery — wash it) as the negative.
- Do not let the plates touch. A cardboard or plastic spacer helps.
- Electrolyte: salt water or vinegar in a jar. Lemon/potato cells work but are weaker.
- One cell is about 0.7–1.1 V. Stack cells in series (copper of one to zinc of the next) for more voltage.
- Current is small and the zinc is consumed. This is stored chemical energy, not free energy.

Do not use lead-acid car-battery acid for a “better” homemade cell unless you already know battery chemistry and ventilation. Hydrogen off-gassing and acid burns are not worth a brighter LED.

5. Water — hydro and wet cells

A. Micro-hydro (motion of water)

Falling or fast water has kinetic and potential energy. A wheel or a small turbine turns a generator (the same copper-coil machine as Chapter 4). Site matters more than gadgets: a creek with drop (head) beats a wide slow river.

Simple demonstration: a plastic water-wheel or Pelton-style spoon wheel on a shaft, a belt or direct coupling to a small DC motor used as a generator, and a hose with pressure from a rain barrel raised a few meters. Measure volts while water hits the cups. Raise the barrel to increase head.

Real off-grid micro-hydro uses a screened intake, a pipe (penstock), a matched turbine, a charge controller, and batteries. Permits, fish screens, and land rights apply. This booklet does not tell you to dam a public stream. If you own a steep seasonal creek, buy a purpose-built micro-hydro kit rather than machining a high-speed runner from scrap.

B. Water as electrolyte only

Distilled water barely conducts. Salt or vinegar in water lets the copper–zinc cell work. That is still a battery. Electrolysis (running current through water to make hydrogen and oxygen) is the reverse: you spend electricity. Burning the hydrogen later cannot give you a surplus. Treat “water fuel” and HHO kits sold as mileage miracles as the same conservation-of-energy failure described in Chapter 1.

Water and electricity

Keep the thermoelectric cold-side water away from live connections. Do not stand in a creek with mains-voltage equipment. For hydro demos, stay at low-voltage DC motors.

6. Other honest tools

Method	What you put in	Typical output
Solar panel	Sunlight	The only “set and forget” source for most people. Use a charge controller.
Bicycle dynamo / hand crank	Your muscle	Phone emergency charge. Exhausting as a lifestyle.
Small wind turbine	Steady wind, a tower, and a dump load	Works on ridges and coasts. Noise, permits, and birds matter.
Thermoelectric (Ch. 3)	Fire plus a cold sink	Watts, not kilowatts.
Micro-hydro (Ch. 5)	Reliable falling water	Best 24-hour off-grid source if you truly have the site.

If the goal is actually useful household energy, a used solar panel, a PWM or MPPT controller, and a deep-cycle battery beat every copper-jar and campfire trick in this book. The fire and copper chapters are for understanding, emergencies, and science — not for hiding the electric bill.

7. What does not work

- **Perpetual-motion magnets.** Magnets have a field, not an infinite fuel tank. Once you extract work, you must put energy back in to keep the motion going.

- **Bedini, overunity, and “radiant energy” kits.** They either run down a battery or measure voltage without measuring current.
- **Earth batteries and “free power from the ground”** beyond a weak electrochemical cell. You are still consuming metal or using a tiny telluric voltage.
- **Running a generator from a motor that the generator powers.** Friction always wins.
- **Fuel from water with a car alternator.** The alternator load on the engine costs more fuel than the hydrogen returns.

A quick check for any viral video: do they show a wattmeter on both input and output for minutes, with the device starting from a dead battery? If not, it is entertainment.

8. A weekend experiment plan

Saturday morning — battery: Three jars, copper, zinc, salt water, series wiring, light an LED. Write down voltage of one cell and of the stack. Weigh the zinc before and after a few hours.

Saturday afternoon — Faraday: Wind a coil, glue magnets to a jar lid, spin with a drill on low or by hand, watch the LED flicker. Add a diode and a capacitor and see the light smooth out.

Sunday — heat: One Peltier module, a mug of ice, a warm pan (not a forge). Measure voltage as the ice melts. That drop is the lesson: the cold sink is half the machine.

If you have a hose and a hill: Couple a toy water wheel to the same coil or a small DC motor. Compare watts (volts $\times$ amps into a resistor) with the battery and the Peltier. Hydro usually wins if the water does not stop.

How to compute a little power

Power in watts $\approx$ volts $\times$ amps. A 2 V cell at 20 mA is 0.04 W. A 5 V Peltier trickle at 200 mA is 1 W. A phone charger wants on the order of 5–10 W. That is why campfire electricity is a demonstration or an emergency trickle, not a lifestyle.

9. Closing notes

Fire gives you a heat difference. Copper gives you a conductor and, with zinc, a battery plate. Water gives you an electrolyte or a turning wheel. Magnets plus motion give you a generator. Put together honestly, they make small, real electricity. Put together as “unlimited free power,” they make disappointed people and sometimes fires.

For a cabin or a phone in a blackout, buy solar and a battery, or a certified generator used outdoors with a transfer switch installed by an electrician. Use this booklet to understand why those things work — and to run a few experiments that actually obey physics.

This text is educational. It is not professional electrical, plumbing, or fire-safety advice. Local law may restrict generators, stream use, and open fires. You are responsible for not burning down a deck, shocking anyone, or connecting anything to building wiring.